Cloud Cost Models

in Amazon Web Services (AWS)

An In-Depth Analysis | Pricing Strategies, Use Cases & Optimization

On-Demand	**Reserved**	**Spot**
Savings Plans	**Dedicated**	**Free Tier**

Agenda

AWS Pricing Overview

Compute

EC2, Lambda, ECS
Priced per hour/second
or by invocation/duration

Storage

S3, EBS, EFS, Glacier
Priced per GB/month
with tiered discounts

Data Transfer

Ingress usually free
Egress priced per GB
Regional transfers cheaper

Fundamental Pricing Principles

- ✓ Pay for what you use — no upfront infrastructure investment required
- ✓ Pay less when you reserve — commit to usage for predictable discounts
- ✓ Pay less with volume — tiered pricing rewards economies of scale
- ✓ Free Tier available — 12-month free usage for 100+ AWS services

01 On-Demand Pricing

What is On-Demand?

Pay for compute/database capacity by the hour or second with no long-term commitments or upfront payments. Ideal for unpredictable workloads.

Billing Granularity	Per second (min 1 min)
Commitment	None — start/stop anytime
Typical Discount	0% (baseline price)

Advantages

- No upfront investment required
- Full flexibility — scale instantly
- No cancellation penalties
- Suitable for short-term, spiky loads
- Best for dev/test environments

Limitations

- Highest per-unit cost of all models
- Not cost-effective for steady workloads
- No financial commitment benefits
- Can lead to cost overruns if unmonitored
- Budget unpredictability at scale

📌 Real-World Use Case

Startup e-commerce platform launching a new product: Traffic is unpredictable during the launch week. On-Demand EC2 instances spin up immediately to absorb the load and are terminated post-launch — avoiding wasted reserved capacity costs.

02 Reserved Instances (RIs)

RI Type	Term	Max Discount	Flexibility	Best For
Standard RI	1 or 3 yr	Up to 72%	Low	Steady-state, same instance type
Convertible RI	1 or 3 yr	Up to 54%	High	When instance family may change
Scheduled RI	1 yr	~5-10%	Med	Recurring capacity needs (daily/weekly)

Payment Options

All Upfront

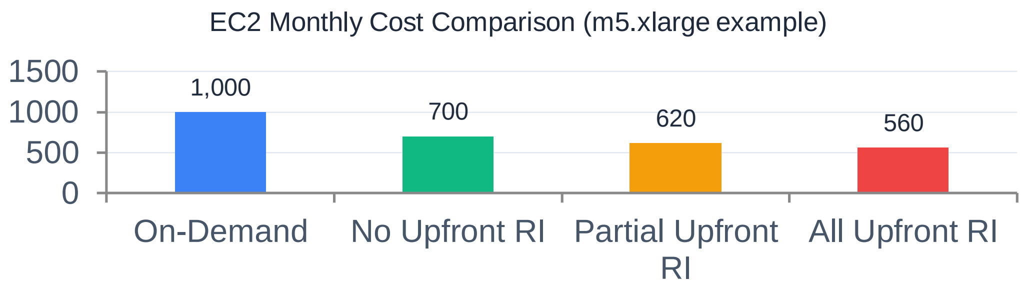
Highest discount, full payment at start

Partial Upfront

Moderate discount, split payment

No Upfront

Lowest discount, monthly billing



📌 Use Case

A SaaS company running MySQL RDS databases 24/7 switches from On-Demand to 3-year All-Upfront Standard RIs, saving ~72% — recouping the upfront cost in under 5 months.

03 Spot Instances

90%
Max Discount
vs On-Demand

2 min
Interruption
Warning

Spare
Uses AWS
Unused Capacity

How Spot Pricing Works

- AWS auctions unused EC2 capacity at Spot prices, which fluctuate with supply/demand.
- You set a maximum bid price; if spot price exceeds it, instance is interrupted.
- AWS gives a 2-minute interruption notice before reclaiming the instance.
- Spot price history is publicly available and shows typical 60–90% savings.
- Spot Fleet + Diversification reduces interruption risk across availability zones.

✓ Suitable Workloads

- Batch data processing
- Machine learning training
- Genomics & HPC
- CI/CD test pipelines
- Video rendering farms

✕ Not Suitable For

- Production web servers
- Real-time financial tx
- Stateful databases
- Long-running interactive
- PCI/HIPAA critical apps

Spot Fleet Architecture — Fault-Tolerant Pattern

Auto Scaling Group → Spot Fleet (Multi-AZ) → SQS Queue (Job Buffer) → On-Demand (Fallback) → S3 / EFS (Results)

📌 Use Case: Netflix batch transcoding — runs 90% of encoding jobs on Spot, saving millions annually. Fallback to On-Demand when interrupted.

04 Savings Plans

Savings Plans offer flexible pricing (up to 66% discount) in exchange for a consistent usage commitment ($/hour) over 1 or 3 years — without locking into specific instance types. Launched in 2019 as a more flexible alternative to RIs.

Up to 66%

Compute Savings Plans

Scope: EC2, Fargate, Lambda
Flexibility: Highest — any instance family, size, OS, region

Up to 72%

EC2 Instance Savings Plans

Scope: EC2 Only
Flexibility: Medium — locked to instance family in one region

Up to 64%

SageMaker Savings Plans

Scope: Amazon SageMaker
Flexibility: Medium — any instance type, component, region

Reserved Instances vs. Savings Plans — At a Glance

Attribute	Reserved Instances	Savings Plans
Commitment Type	Specific instance type/region	$/hour spend commitment
Instance Flexibility	Limited (Convertible RI only)	High (Compute SP)
Applies to Fargate/Lambda	❌ No	✅ Yes (Compute SP)
Max Discount	72%	66%

📌 Use Case: A fintech firm running containerized microservices on Fargate commits $500/hr via Compute Savings Plans — saving 66% vs On-Demand, while retaining freedom to change instance families as their architecture evolves.

05 & 06 Dedicated Hosts & Free Tier

Dedicated Hosts

- Physical server dedicated to your account — no other tenants share hardware
- Supports Bring Your Own License (BYOL) for Windows Server, SQL Server, Oracle
- Required for certain compliance mandates: PCI-DSS, HIPAA, FedRAMP
- Pricing: On-Demand Dedicated ($3–$6/hr) or Reserved Dedicated (up to 70% off)
- Dedicated Instances (vs Hosts): Dedicated hardware but AWS manages placement

📌 Use Case: Healthcare analytics company processing patient records needs HIPAA-compliant isolated hardware. Dedicated Hosts with BYOL Oracle DB reduces licensing cost by 40% vs standard On-Demand.

AWS Free Tier

12-Month Free

EC2: 750 hrs/mo (t2.micro) • S3: 5 GB storage • RDS: 750 hrs/mo (db.t2.micro) • CloudFront: 50 GB data transfer

Always Free

Lambda: 1M requests/mo • DynamoDB: 25 GB storage • SNS: 1M publishes/mo • SES: 62K emails/mo

Short-Term Trial

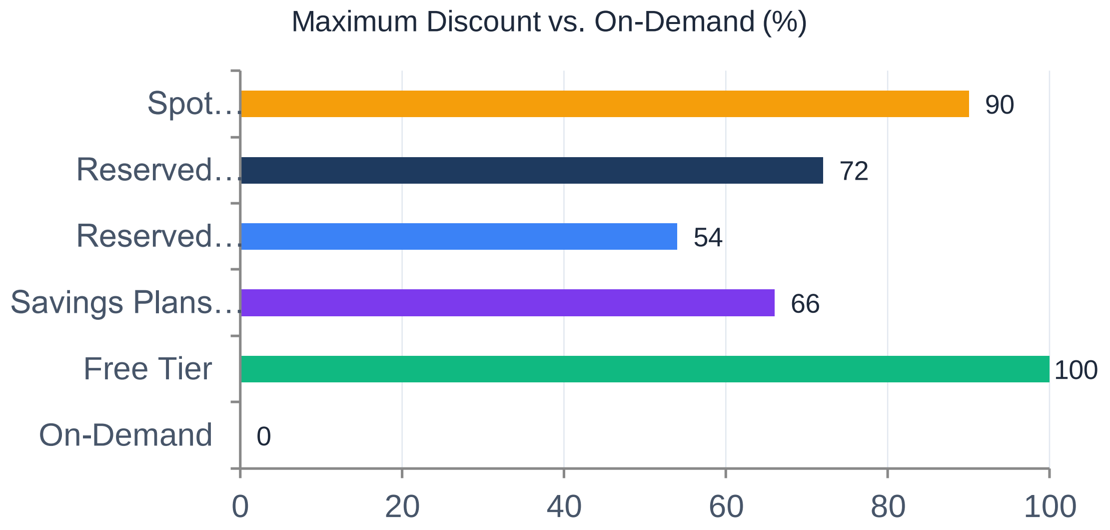
SageMaker: 2 months • Redshift: 2 months • AWS Detective: 30 days • Macie: 30 days

📌 Use Case: An independent developer builds and validates a full-stack app — EC2 + S3 + RDS + Lambda — entirely within the Free Tier for 12 months, achieving $0 infrastructure cost while validating product-market fit.

⚡ Data Transfer Pricing — Often Overlooked

Inbound data to AWS: FREE | Outbound first 100 GB/mo: FREE | Outbound 1–10 TB: $0.09/GB | Cross-AZ transfer: $0.01/GB (each direction) | Same-region transfer: $0.00–$0.01/GB | Use VPC Endpoints to avoid data transfer fees for S3/DynamoDB.

Comprehensive Cost Comparison & Analysis



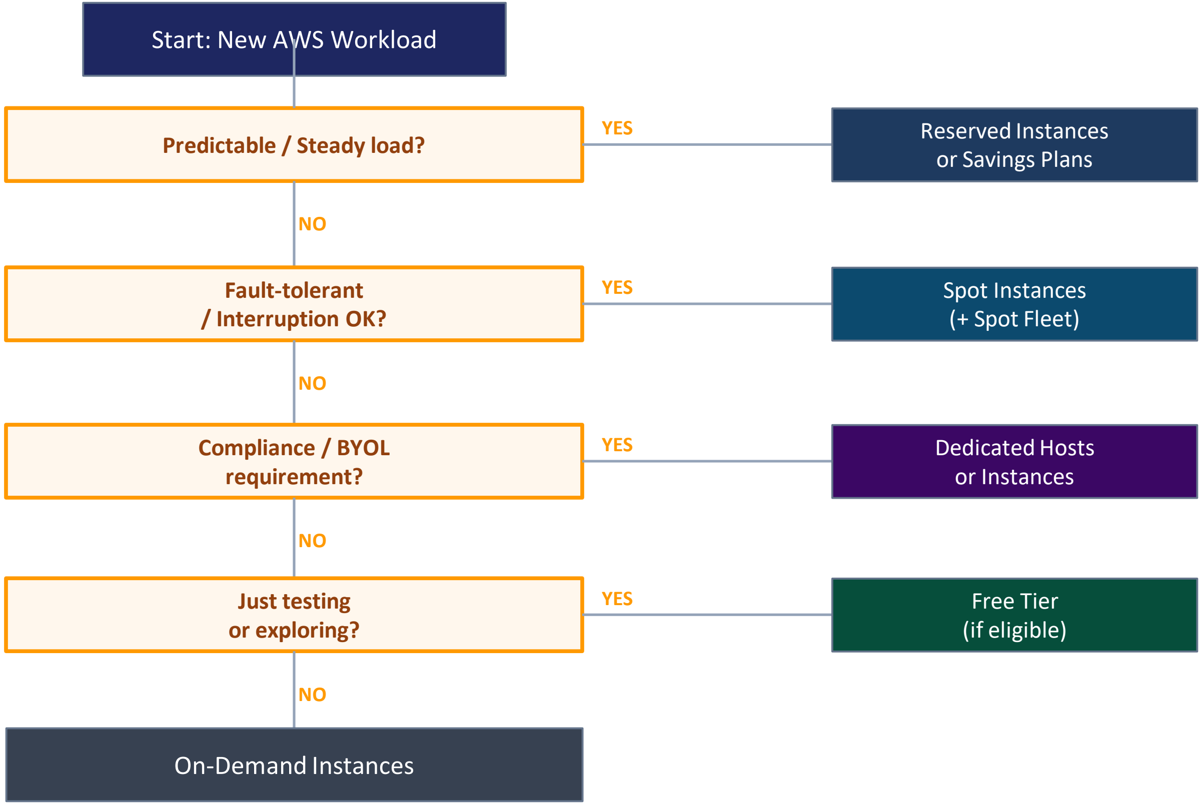
Decision Matrix

Model	Discount	Commitment
On-Demand	0%	None
Reserved	72%	1-3 yr
Spot	90%	None
Savings Plans	66%	1-3 yr
Dedicated	0-70%	Optional
Free Tier	100%	None

Cost Optimization Best Practices

1. Use AWS Cost Explorer to analyze usage patterns before committing to RIs
2. Combine Spot + On-Demand in Auto Scaling for resilient cost optimization
3. Right-size instances with Compute Optimizer recommendations
4. Enable S3 Intelligent-Tiering for automatic cost reduction on infrequently accessed data
5. Schedule non-prod instances to stop during off-hours (saves 65%+ of compute costs)
6. Use Reserved Capacity for databases; Savings Plans for compute; Spot for ML training

Cost Model Decision Framework



Tip: Most production workloads benefit from a blended strategy — On-Demand for burst, Reserved/Savings Plans for baseline, Spot for batch.

Real-World Use Cases Across Industries

Media & Entertainment

Spot + On-Demand

e.g., Netflix-style OTT

Video encoding/transcoding workloads run on Spot (90% cost saving). Live streaming uses On-Demand for guaranteed availability.

💰 ~78% cost reduction on encoding

Healthcare

Dedicated + Reserved

e.g., Hospital EHR system

HIPAA-compliant patient data processing on Dedicated Hosts. Predictable EMR database workloads on 3-yr Standard RIs.

💰 HIPAA compliance + 65% savings

E-Commerce

On-Demand + Reserved

e.g., Retail Platform

Reserved Instances handle baseline traffic. On-Demand scales up for seasonal peaks (Black Friday, Diwali sales).

💰 55% saving on stable load

Financial Services

Savings Plans + On-Demand

e.g., Algorithmic Trading

Compute Savings Plans for consistent risk model computation. On-Demand for burst capacity during volatile market hours.

💰 60% saving on compute costs

Research & Academia

Spot Instances

e.g., Genomics Lab

Genomic sequencing analysis on Spot Fleet across multiple AZs. Checkpoint-based jobs restart seamlessly after interruptions.

💰 85% cost reduction vs on-prem

Startups / SaaS

Free Tier → On-Demand

e.g., Early-Stage SaaS

12-month Free Tier for MVP development. Migrate to On-Demand as product launches, then Reserved once traffic patterns are clear.

💰 $0 infrastructure in Year 1

Key Takeaways

01 No single pricing model fits all — AWS cost optimization requires a blended, workload-aware strategy.

02 On-Demand is the baseline. Move workloads to Reserved/Savings Plans once usage patterns stabilize.

03 Spot Instances unlock the highest savings (90%) but require fault-tolerant architecture patterns.

04 Savings Plans are the modern evolution of RIs — more flexible, applicable to Fargate and Lambda.

05 Always model TCO including data transfer, storage tiers, and licensing — not just compute costs.

06 Use AWS Cost Explorer, Compute Optimizer, and Trusted Advisor to continuously right-size and optimize.